

MUHAMMAD MUSTAFA SHAH

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Education

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology

Sep 2022 – Present

Bachelor of Science in Data Science — CGPA: 3.15 (Dean's Honour List)

Topi, Pakistan

- Coursework:** Generative AI & LLMs, Big Data Analytics (PySpark), MLOps, Data Warehousing, Deep Learning, Software Engineering.

Professional Experience

BIOMISA Lab

Aug 2025 – Sep 2025

Research Assistant - Data Analytics

NUST, Pakistan

- Engineered clinical risk models for Physical Frailty Assessment, processing 50+ biomarker datasets.
- Optimized feature extraction using Pandas/NumPy vectorization, reducing preprocessing latency by 40%.
- Standardized unstructured patient records into FHIR-compliant formats, achieving 95% data integrity.

ITSOLERA PVT LTD

June 2024 – Aug 2024

Machine Learning Intern (Social Data Analysis)

Remote

- Fine-tuned RoBERTa transformers for sentiment classification on election data, achieving 85% F1-score.
- Built scalable ETL workflows processing 100GB+ of social streams using SQL & Python.
- Deployed inference endpoints on Power BI dashboards, optimizing query response to < 1s.

CodexCue

June 2024 – July 2024

Data Analytics Intern

Remote

- Performed Hypothesis Testing on metrics of 20+ institutions to identify statistically significant disparity gaps.
- Automated policy reporting using Python scripting, eliminating 15+ hours of manual data entry weekly.

Projects (AI & Digital Innovation)

SpeechEcho: Zero-Shot Voice Cloning (Final Year Project) | PyTorch, Chatterbox, Docker

Present

- Architected a zero-shot TTS system using Chatterbox to clone voices from 3-second audio prompts.
- Integrated VQ-VAE for acoustic tokenization, reducing GPU inference latency to sub-200ms.
- Deployed Dockerized FastAPI endpoints to serve concurrent synthesis requests with 99.9% uptime.

Multimodal Agentic Media Pipeline | Whisper Large-v3, Gemini 1.5 Pro, LangChain

Feb 2025

- Orchestrated an agentic workflow using Whisper and Gemini 1.5 for automated semantic extraction.
- Engineered a serverless pipeline processing 50+ hours daily, cutting localization costs by 90%.
- Implemented Multimodal RAG to ground outputs in source audio, eliminating hallucinations.

Pakistan Stock Analysis & Forecasting (FinGPT) | Python, FinGPT, LSTMs

July 2024

- Developed a forecasting engine for PSX market trends using FinGPT and LSTMs.
- Processed 5 years of high-frequency data, improving prediction accuracy by 15% over baselines.
- Built an automated dashboard visualizing market volatility and investor sentiment.

Research Paper Companion (High-Performance RAG) | Llama-3, FAISS, Sentence-BERT

Dec 2025

- Built a high-precision RAG pipeline for academic queries using Sentence-BERT embeddings.
- Engineered a FAISS backend enabling semantic search over 10k+ chunks with < 50ms latency.
- Optimized retrieval parameters for Llama-3, maximizing context relevance and accuracy.

Technical Skills

AI & Deep Learning: Multimodal Agents, Deep Learning (CNNs, Transformers), LLMs (RAG, Fine-tuning), Whisper AI

Data Engineering: AWS (Lambda, EC2), PySpark, Hadoop, SQL, ETL Pipelines, Docker, Git

Data Visualization: Microsoft Power BI, Tableau, Matplotlib, Seaborn

Programming: Python (Pandas, NumPy, Scikit-learn), C/C++, MySQL, React, FastAPI

Soft Skills: NGO Management, Public Speaking, Strategic Planning, Cross-Cultural Communication

Languages: English (Fluent), Urdu, Pashto (Native)

Social Impact & Leadership

Amal Academy (Career-Prep Fellowship)

Feb 2025 – May 2025

Fellow & Project Lead (Project Udhaan)

Pakistan

- Co-founded a social initiative connecting industry experts with underprivileged high school students for mentorship.

Project Topi (GIKI Welfare Society)

Dec 2022 – Present

Society Head (Best Dedication Award)

GIKI

- Spearheaded fundraising and digital outreach for a student-run NGO, managing resource allocation for welfare.